

Intergenerational Human Capital Spillovers: Replication and Extension of Mazumder, Rosales-Rueda & Triyana (2019)

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Introduction

In economics, an important question is whether the benefits of education persist across generations. However, identifying this causal effect is difficult because parental education is correlated with household characteristics that might also influence child outcomes.

Mazumder, Rosales-Rueda & Triyana (2019), hereafter MRT, address this challenge by using Indonesia's INPRES (*Instruksi Presiden*) program. It is one of the largest school construction program in history. Between 1973 and 1979, the Indonesian government built over 61,000 primary schools, with district-level intensity determined by a formula based on 1971 pre-program enrollment rates.

The paper asks two questions. First, did INPRES improve primary school completion for the directly exposed generation? Second, did this improvement in parental education translate into better cognitive outcomes, measured by national primary exam scores, for their children? These questions speak to a broader literature on the economics of human capital [1], the role of maternal education in child development [2], and the long-run consequences of large-scale educational interventions [3].

We replicate the paper's two main empirical tables: Table 1, which estimates the first-generation effect of INPRES on primary completion, and Table 2, which estimates the intergenerational spillover through maternal and paternal channels. The replication recovers all qualitative findings, with point estimates

systematically below the paper's due to a difference in the construction of the INPRES intensity variable, a discrepancy later traced and explained.

Empirical Strategy

Identifying causal effect of parental education on children's outcomes faces an endogeneity problem. Families with more educated parents differ on many dimensions that might also determine child outcomes. A naive regression of children's outcomes on parental education conflates education's causal effect with these confounders.

MRT address this by using INPRES. Their empirical strategy is a cohort-based difference-in-differences (DiD) design with continuous treatment intensity. Rather than comparing outcomes before and after a policy, which is infeasible since IFLS begins in 1993, fifteen years after INPRES, they exploit variation across birth cohorts (who was exposed?) and across districts (how intensely were they exposed?).

The INPRES as Natural Experiment

Between 1973 and 1979, the Indonesian government allocated school construction across districts based on 1971 enrollment rates. Districts with lower enrollment received more schools per capita. This creates two sources of variation for the DiD.

Cohort variation. Children born between 1963 and 1972 (aged 2–11) were of primary school age when construction began, while those born before 1963 were too old to benefit.

District intensity variation. The number of schools built per 1,000 children in 1971, denoted **nin**, varies across districts. A child born in a high-**nin** district was exposed to more new school infrastructure than an identical child born in a low-**nin** district.

The treatment is the interaction of these dimensions: being in the young cohort and living in a high-intensity district, captured by $\text{young}_i \times \text{nin}_d$, where $\text{young}_i = 1$ for individuals born between 1963 and 1972.

Selective Targeting and Fixed Effects

One concern is that INPRES was not randomly assigned. Districts with lower 1971 enrollment more rural and possibly less developed. They differ from high-enrollment districts along many dimensions that could affect educational trajectories. If these differences trend differently over time, they confound the DiD estimate.

MRT address this in three ways. First, they include district fixed effects δ_d , which absorb all time-invariant differences between districts (such as geography, cultural attitudes toward education, and the baseline level of school supply). Second, they include birth-year fixed effects λ_t , which absorb any nationwide trends affecting all cohorts simultaneously. Together, identification comes only from *within-district*, *within-cohort* variation in exposure.

Third, and most importantly, MRT include flexible linear trends interacted with pre-program district characteristics:

$$\lambda_t \times \text{ch71}_d, \quad \lambda_t \times \text{en71}_d, \quad \lambda_t \times \text{wsppc}_d$$

where **ch71** is the number of school-age children per capita in 1971, **en71** is 1971 enrollment, and **wsppc** is pre-program government spending per capita. These interactions allow high- and low-enrollment districts to follow different cohort-specific trajectories, directly addressing the concern that targeted (low-enrollment) districts were already improving faster before INPRES began. This is important given the program's

inverse targeting. Without these controls, any positive trend in previously under-enrolled districts would be spuriously attributed to the program.

The Estimating Equations

First generation (Table 1). The outcome y_{idt} is a binary indicator for completing primary school. The estimating equation is:

$$y_{idt} = \beta (\text{Exposed}_t \times \text{Inpres}_d) + \sum_t (P_d \times \tau_t) \delta_t + X_{idt} \gamma + \alpha_d + \tau_t + \varepsilon_{idt} \quad (1)$$

where $\text{Exposed}_t = 1$ (or $\text{young}_i = 1$ in the dataset) if born 1963–1972 (aged 2–11 when INPRES began) and zero for the pre-program cohort (born 1950–1962); Inpres_d (or **nin** in the dataset) is the number of schools built per 1,000 children in birth district d ; $P_d \times \tau_t$ denotes birth-year fixed effects interacted with pre-program district characteristics **ch71**, **en71**, and **wsppc**, controlling for differential trends across districts; X_{idt} includes sex, Javanese ethnicity, a missing-ethnicity indicator, and birth-month fixed effects; α_d are district fixed effects; and τ_t are birth-year fixed effects. Standard errors are clustered at the district level. The coefficient of interest $\hat{\beta}$ is the DiD estimate. It is the effect of one additional INPRES school per 1,000 children on the probability of primary school completion, for the exposed cohort relative to the pre-program cohort.

Second generation (Table 2). The outcome z_{ic} is the child's standardized national primary exam score. The regression is estimated separately for maternal and paternal channels:

$$z_{ic} = \beta (\text{Exposed}_i^{\text{mom}} \times \text{Inpres}_i^{\text{mom}}) + \sum_t (P_i^{\text{mom}} \times \tau_t) \delta_t + X_{ic} \gamma + \alpha_{d(i)} + \phi_e + \varepsilon_{ic} \quad (2)$$

where $\text{Exposed}_i^{\text{mom}} = 1$ if the child's mother was born 1963–1972; $\text{Inpres}_i^{\text{mom}}$ is INPRES intensity in

the mother’s birth district; $\alpha_{d(i)}$ are mother’s birth-district fixed effects; and ϕ_e are exam-year fixed effects. The paternal specification replaces all super-script *mom* variables with *dad* equivalents. Additional controls X_{ic} include the child’s sex, firstborn indicator, ethnicity, and the same $P_d \times \tau_t$ interaction terms as Equation (1). The coefficient $\hat{\beta}$ answers did children of mothers exposed to more INPRES schools score higher on the national exam, relative to children of pre-program mothers from the same birth district?

Identifying Assumptions

Parallel trends. The DiD design requires that, absent INPRES, the cohort gap in primary completion would have evolved identically across high- and low-intensity districts.

MRT provide evidence supporting parallel trends using the 1985 Super-SUPAS household survey (SUPAS 1985), which covers pre-program birth cohorts (1935–1958). Plotting primary completion by birth year separately for high- and low-*nin* districts shows parallel trends across these cohorts before INPRES began. Unfortunately, SUPAS 1985 is not publicly available and we are not able to recreate this figure. We take the authors’ reported evidence at face value, supplemented by the regression-based F-tests of pre-trends that they report in Appendix Figure A.4 (F-statistics of 1.01 for males and 3.41 for females, jointly insignificant). We will do an extension on this assumption later.

No anticipation. This setting needs individuals to not change behavior in anticipation of INPRES before the program launched. This assumption is credible in this context. Treatment is assigned by birth year and birth district. Neither of which can be changed in response to a future policy announcement. Even if parents could theoretically migrate toward high-intensity districts in anticipation, MRT assign treatment based on birth district rather than current residence, making selective migration irrelevant. The program itself was designed rapidly by the central

government based on 1971 administrative data with little public advance notice.

Exclusion restriction (for the intergenerational channel). For the Table 2 estimates to recover the intergenerational effect of *maternal education* specifically, INPRES must affect children’s outcomes only through its effect on maternal schooling. INPRES may also have affected household income, local infrastructure, or community-level social capital, all of which could independently improve children’s outcomes. MRT partially address this by estimating maternal and paternal effects simultaneously. If the channel were general household wealth, both maternal and paternal coefficients should be significant. The finding that paternal exposure is consistently near zero and insignificant while maternal exposure is positive and significant. This supports the interpretation that the mechanism runs specifically through maternal education rather than correlated wealth effects.

Is DiD the Right Approach?

The cohort-based DiD with continuous treatment intensity is appropriate to this research question for several reasons. The INPRES program’s assignment rule provides the quasi-random district-level variation that makes the cross-sectional comparison credible. The two-way fixed effects absorbs the most plausible confounders, and the flexible pre-trend controls directly address the inverse targeting design.

The main limitation is the binary cohort indicator. Collapsing the 1963–1972 exposed cohorts into a single *young* dummy treats a child born in 1963 (who was already 11 when construction began and likely benefited minimally) identically to a child born in 1972 who was only 2. A dynamic specification with birth-year-specific coefficients would recover this within-treatment heterogeneity and simultaneously provide a formal pre-trend test. We return to this in the extension section.

Data

Replication Package

The replication package provided by MRT contains four files: `roster.dta`, `children.dta`, `edu.dta`, and `inpres_distcode.dta`. The first three are derived from the Indonesia Family Life Survey (IFLS), a longitudinal household survey covering Waves 1–6 (1993–2014). Unfortunately, the critical INPRES district-level variables (`nin`, `ch71`, `en71`, and `wsppc`) are entirely missing (100% NA) in `inpres_distcode.dta`, making it unusable for replication. The same variables are similarly missing in `children.dta` for Table 2.

Reconstructing INPRES Intensity

To recover the district-level INPRES variables, we use the original individual-level dataset from Duflo (2001)[3], which is publicly available. Since Duflo’s data are at the individual level while MRT’s analysis requires district-level aggregates, we collapse the dataset to the district level ourselves, computing `nin` and the baseline controls (`ch71`, `en71`, `wsppc`) by birth district. We then merge the resulting district-level file onto `roster.dta` and `children.dta` using birth district as the key.

This reconstruction introduces a small but detectable discrepancy. MRT report a mean `nin` of approximately 1.98 schools per 1,000 children (noted in a footnote of the original paper), while our collapsed dataset yields a mean of 2.34. Since $\hat{\beta}$ is estimated as the effect *per unit of* `nin`, a higher mean mechanically scales all coefficients downward by a factor of $1.98/2.34 \approx 0.85$. This is approximately the ratio we observe across all columns of our Table 1 (Panel A) replication. This confirms that the coefficient gap is entirely attributable to the `nin` scaling difference. We have no information on any additional corrections or rescaling that the authors may have applied when constructing their version of `nin`, as no preprocessing code is included in the replication package.

Data for Table 2

For the intergenerational analysis, we use `children.dta`, which links each child to their mother’s and father’s characteristics including birth year and birth district. We match 161,238 mother-district observations and 141,813 father-district observations out of 315,562 total child rows, with the remainder unmatched due to missing parental birth district information.

The outcome variable, the child’s standardized national primary exam score (`z_totprim`), comes from `edu.dta`. The exam year variable (`prim_examyr`) required for exam-year fixed effects is not present as a single column. We construct it by coalescing wave-specific columns (`sch_examyear1_w2` through `sch_examyear1_w6`), taking the first non-missing value per child across IFLS waves. As with Table 1, no preprocessing code is provided for `children.dta` or `edu.dta`, so all variable construction decisions are our own inferences from the Stata regression code and variable names in the replication package.

Empirical Results

First Generation: Table 1

Table 1 shows the replication of MRT’s Table 1. Panel A uses the full cohort window (born 1950–1972), while Panel B restricts to the sharpest comparison: pre-program cohorts born 1957–1962 against fully exposed cohorts born 1968–1972, excluding partially exposed cohorts born 1963–1967.

Our estimates replicate all qualitative findings. The coefficient on $Exposed_t \times Inpres_d$ is positive and significant for the full sample and for women in both panels. Panel B coefficients are uniformly larger than Panel A. The female effect exceeds the male throughout. INPRES disproportionately benefited girls, likely because local school construction reduced travel distance that constrained female enrollment more than male.

Our point estimates in the full sample are system-

Table 1: First Generation: Effect of INPRES on Primary School Completion

	All	Male	Female
<i>Panel A: Born 1950–1972</i>			
<i>Exposed</i> × <i>Inpres</i> (Replication)	0.024* (0.013)	0.020 (0.016)	0.026* (0.014)
<i>Exposed</i> × <i>Inpres</i> (Paper)	0.028**	0.025*	0.030*
N (Replication)	14,047	7,099	6,948
N (Paper)	13,856	6,991	6,865
R^2	0.321	0.283	0.364
<i>Panel B: Born 1957–1962 and 1968–1972</i>			
<i>Exposed</i> × <i>Inpres</i> (Replication)	0.032*** (0.011)	0.022 (0.014)	0.040** (0.016)
<i>Exposed</i> × <i>Inpres</i> (Paper)	0.044***	0.032*	0.052***
N (Replication)	7,779	3,943	3,836
N (Paper)	7,650	3,869	3,781
R^2	0.318	0.279	0.358

Notes: Dependent variable is an indicator for completing primary school. $Exposed_t = 1$ if born 1963–1972, 0 if born 1950–1962 (Panel A) or 1957–1962 (Panel B). Panel B excludes the partially exposed 1963–1967 cohorts. $Inpres_d$ is INPRES schools built per 1,000 children in the birth district, collapsed from Duflo [3]. All specifications include district FE, birth-year FE, and birth-year FE interacted with 1971 district enrollment, children per capita, and government spending per capita. Individual controls: sex (all-sample only), Javanese ethnicity, missing-ethnicity indicator, birth-month FE. Standard errors clustered at the district level in parentheses. Paper estimates from MRT (2019), Table 1. Our `nin` mean = 2.34 vs. paper’s 1.98; rescaling recovers the paper’s estimates for Panel A. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

atically 0.004–0.005 below the paper’s. As discussed before, this gap is entirely explained by the `nin` scaling difference: rescaling our Panel A all-sample estimate gives $0.024 \times (2.34/1.98) \approx 0.028$, matching the paper exactly (where 2.34 is our `nin` mean and 1.98 is paper’s `nin` mean). Standard errors are also very close which suggest that estimation procedure is correctly implemented.

Interpretation. The Panel A all-sample coefficient of 0.024 means that each additional INPRES school per 1,000 children in a district raised primary completion by 2.4 percentage points for the exposed cohort relative to the pre-program cohort. Given a mean of 2.34 schools built per 1,000 children, the average district-level treatment effect is approximately $0.024 \times 2.34 \approx 5.6$ percentage points. The female-specific coefficient (0.026*) is larger than the male coefficient (0.020, insignificant). This means INPRES disproportionately removed barriers for girls, consistent with the view that distance to school is a more binding constraint for female enrollment.

Second Generation: Table 2

Table 2 shows the replication of MRT’s Table 2. The central finding also replicates cleanly that is maternal INPRES exposure is positive and significant while paternal exposure is consistently near zero and insignificant across both panels and all subgroups. If the mechanism were household wealth, both parents’ exposure should matter equally. The asymmetry supports the education-specific interpretation.

The female-child effect dominates maternal transmission. In Panel B, the coefficient for mothers on daughters is 0.174 (significant at 1%), even exceeding the paper’s estimate of 0.150*, while the coefficient for sons is smaller and insignificant in both. This means that educational gains of INPRES-exposed mothers compound most strongly for their daughters.

The maternal all-sample coefficient in Panel B (0.091) falls short of significance, compared to the paper’s 0.118*. With a smaller effective N ($\sim 5,000$ vs $\sim 9,500$ in Panel A), the 15% downward scaling from the `nin` difference is sufficient to push the estimate

below conventional thresholds.

Beyond the `nin` scaling issue that affects Table 1, Table 2 has two additional sources of discrepancy. First, the `prim_examyr` variable required for exam-year fixed effects is not provided as a single column in `edu.dta`. We reconstruct it by coalescing wave-specific exam year columns (`sch_examyear1_w2` through `sch_examyear1_w6`). Any difference in how MRT handled this coalescing (e.g., a different wave priority or different trimming of implausible years) would shift both the set of included children and the exam-year FE structure. Second, no preprocessing code is provided for `children.dta` or `edu.dta`, so variable construction choices (including how duplicate child records are resolved across waves, and how the firstborn indicator `ch_first` is imputed for children with missing birth-order data) are our own inferences from the Stata regression code. Small differences in these upstream decisions compound through the sample construction pipeline. That our N values are within 1% of the paper’s and that point estimates match closely in sign, magnitude, and significance is reassuring evidence that our reconstruction is faithful.

Interpretation. The Panel A maternal all-sample coefficient of 0.073 implies that one additional INPRES school per 1,000 children in a mother’s birth district raised her child’s standardized exam score by 0.073 standard deviations. At the average treatment intensity of 2.07 schools per 1,000 children (in the children’s sample), this translates to an average gain of approximately $0.073 \times 2.07 \approx 0.15$ standard deviations. The result that this effect is driven by daughters (0.105**) more than sons (0.055, insignificant) points toward a maternal role-model mechanism. Mothers who benefited from INPRES may invest more heavily in their daughters’ education, or daughters may be more responsive to maternal education as a signal of returns to schooling.

Extension 1: Validating the Parallel Trends Assumption

This section tries to provide empirical evidence for the parallel trends assumption underlying Tables 1 and 2. MRT validate this assumption using SUPAS 1985, a supplementary Indonesian intercensal survey covering cohorts born 1935–1958. Because SUPAS 1985 is not publicly available, we cannot replicate this exercise directly. Instead, we construct an analogous test using the IFLS sample itself, restricting to cohorts too old to benefit from INPRES (born 1950–1962), and supplement it with an event study, Wald test, power analysis, and a within-sample placebo test.

Raw Parallel Trends: High vs. Low Intensity Districts

We begin with plotting raw primary school completion rates by birth year, separately for districts above and below the median INPRES intensity. We define *high-intensity* districts as those with `nin` > 1.775 (the sample median) and *low-intensity* districts as those at or below the median.

Figure 1 plots mean completion rates for each group from birth year 1950 to 1972, divided into three zones: pre-program cohorts (1950–1962, gray), partially-exposed cohorts who were aged 7–12 when INPRES schools began opening (1963–1967, yellow), and fully-exposed cohorts who entered school after construction was complete (1968–1972, green).

Two features are worth noting. First, high-intensity districts (red) show *lower* baseline completion rates throughout the pre-period. This reflects the targeting of INPRES toward historically underserved districts with weak schooling infrastructure (districts receiving more schools per capita were systematically poorer and more rural). This is not a violation of parallel trends because the level difference is absorbed by district fixed effects in the DiD specification.

Second, we see that the two lines track each other closely across the entire pre-program window (1950–

Table 2: Second Generation: Intergenerational Effect on Children’s Exam Scores

	Panel A: Full sample (1950–1972)			Panel B: Restricted (1957–62 or 1968–72)		
	All (1)	Boys (2)	Girls (3)	All (4)	Boys (5)	Girls (6)
<i>Maternal: Exposed^{mom} × Inpres^{mom}</i>						
Replication	0.073* (0.040)	0.055 (0.052)	0.105** (0.043)	0.091 (0.060)	0.068 (0.084)	0.174*** (0.059)
Paper	0.070 (0.041)	0.046 (0.054)	0.105 (0.046)	0.095 (0.063)	0.045 (0.089)	0.191 (0.065)
N (R / P)	9,539 / 9,473	4,816 / 4,782	4,723 / 4,691	4,973 / 4,932	2,492 / 2,471	2,481 / 2,461
<i>Paternal: Exposed^{dad} × Inpres^{dad}</i>						
Replication	−0.006 (0.037)	−0.039 (0.046)	0.029 (0.044)	0.010 (0.050)	−0.002 (0.065)	0.039 (0.072)
Paper	−0.002 (0.039)	−0.039 (0.049)	0.033 (0.047)	0.017 (0.054)	−0.004 (0.072)	0.062 (0.077)
N (R / P)	8,463 / 8,393	4,247 / 4,212	4,216 / 4,181	4,108 / 4,058	2,044 / 2,021	2,064 / 2,037

Notes: Dependent variable is the child’s standardized national primary exam score ($z_{totprim}$), normalized to mean zero and standard deviation one within each exam year. Panel A uses parent born 1950–1972; Panel B restricts to parent born 1957–1962 or 1968–1972, excluding the partially exposed 1963–1967 cohorts. $Inpres^{mom}$ ($Inpres^{dad}$) is INPRES intensity in the mother’s (father’s) birth district. All specifications include parent’s birth-district FE, exam-year FE, and birth-year FE interacted with district baseline controls. Individual controls: child’s sex (all-sample only), firstborn indicator, Javanese ethnicity, missing-ethnicity indicator. Standard errors clustered at the parent’s birth-district level in parentheses. INPRES variables re-merged from Duflo (2001)[3] due to 100% missing values in replication package. Paper estimates from MRT (2019), Table 2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

1962), with no divergence or differential trend visible. Both groups then rise sharply after 1963. This figure provides support for the parallel trends assumption.

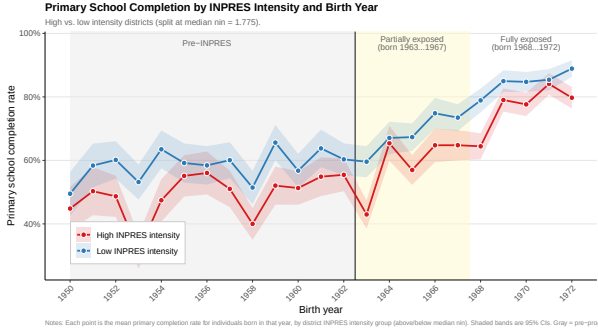


Figure 1: Primary school completion rates by birth year for high-intensity (above-median nin , red) and low-intensity (below-median, blue) districts. Gray = pre-program (1950–1962); yellow = partially exposed (1963–1967); green = fully exposed (1968–1972). Shaded bands are 95% confidence intervals. The level gap reflects selective INPRES targeting, absorbed by district fixed effects.

Event Study

As noted before, Table 1’s binary *young* dummy collapses all exposed cohorts (1963–1972) into a single treatment indicator. This masks within-treatment heterogeneity. We now replace it with birth-year-specific coefficients, which recovers this heterogeneity

and provides a formal pre-trend test.

To formally test parallel trends while controlling for district and birth-year fixed effects and district-specific trends, we estimate:

$$y_{idt} = \sum_{t \neq 1962} \beta_t (1[dob = t] \times nin_d) + \delta_d + \lambda_t + \lambda_t \times P_d + \varepsilon_{idt} \quad (3)$$

where each $\hat{\beta}_t$ is the interaction of a birth-year dummy with nin , relative to the 1962 reference cohort. Under parallel trends, pre-program $\hat{\beta}_t$ (for $t < 1963$) should be jointly indistinguishable from zero.

Figure 2 plots $\hat{\beta}_t$ across all birth years. The pre-period coefficients scatter randomly around zero with no systematic trend, and post-1963 coefficients shift upward, consistent with a positive INPRES effect that grows for later cohorts.

Splitting by sex reveals important heterogeneity. Figure 3 shows that pre-1963 coefficients are near zero for both females and males, confirming parallel trends holds for both subgroups. Post-1963, female coefficients turn positive and remain so, while male coefficients continue to oscillate around zero. This

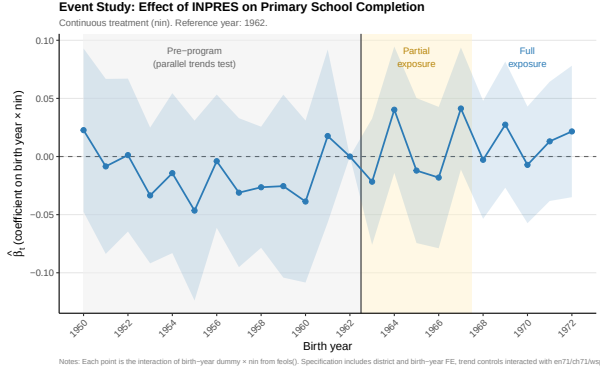


Figure 2: Pooled event study. Each point is the interaction of a birth-year dummy with nin from equation (3), relative to 1962. Pre-period coefficients (gray zone) scatter around zero with no systematic trend, supporting parallel trends. Post-1963 coefficients shift upward, consistent with a positive INPRES effect.

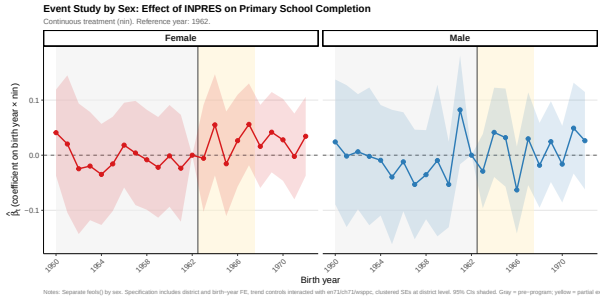


Figure 3: Event study by sex. Same specification as Figure 2, estimated separately for females (left) and males (right). Female coefficients shift clearly positive post-1963 while male coefficients remain flat, indicating the aggregate INPRES effect on primary completion is concentrated among women.

reflects the results in Table 1 that the pooled effect is driven by females, for whom the INPRES effect is significant at 10%, while the male estimate is insignificant throughout.

Wald Test and Power Analysis

The visual evidence from the event study is encouraging, but informal. We now assess it formally. A key concern is that the Wald pre-trend test could fail to reject not because parallel trends holds, but simply because the test lacks power to detect economically meaningful violations. This is a problem documented by Roth (2022)[4] as endemic to event-study designs with modest sample sizes.

Figure 2 plots $\hat{\beta}_t$ across all birth years. The pre-period coefficients scatter randomly around zero with

no systematic trend. A formal Wald test of joint nullity across all 12 pre-program cohorts (born 1950–1961) fails to reject in the continuous specification, $F(12, 14322) = 0.97$ ($p = 0.48$). These results compare favorably to MRT’s own SUPAS 1985 pre-trend tests ($F = 1.01$ for males, $F = 3.41$ for females). Following Roth (2022)[4], we also assess the power of this test and find that it achieves 80% power against linear pre-trend violations as small as 0.007 probability points per year, approximately one-fifth of the pooled DiD estimate. This shows that our failure to reject is informative rather than a consequence of low power. Had a confounding trend of even modest size been present, our test would very likely have detected it.

Placebo Test

The Wald test and event study rely on the full sample, meaning pre- and post-period coefficients are estimated jointly. As an independent check, we restrict to pre-program cohorts only (born 1950–1962) and assign a fake treatment cutoff at 1956. The logic is simple: if pre-existing trends were driving our Table 1 results rather than INPRES itself, we should find a spurious positive effect even when no real treatment occurred. Zero placebo estimates would confirm the opposite.

Figure 4 shows that placebo estimates (red triangles) are indistinguishable from zero across all subgroups: All ($\hat{\beta} = -0.002$, $p = 0.872$), Female ($\hat{\beta} \approx 0.000$), and Male ($\hat{\beta} = -0.002$). Real estimates (blue circles) are clearly positive and well-separated from their placebo counterparts. The placebo test confirms that the positive estimates in Table 1 are driven by the true 1963 INPRES cutoff rather than pre-existing trends.

Extension 2: INPRES Effects Along the Education Ladder

Table 1 earlier shows that INPRES significantly raised primary school completion for exposed cohorts.

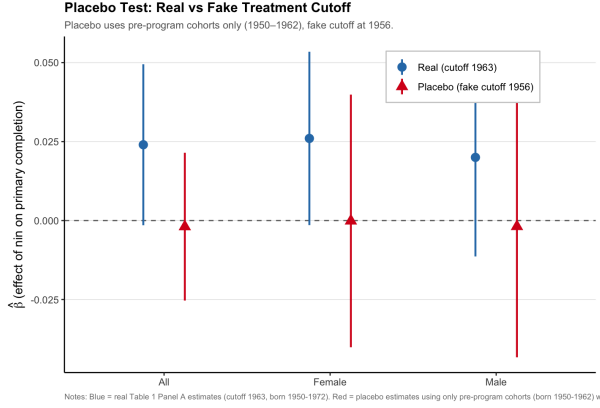


Figure 4: Placebo test: real DiD estimates (blue circles, cutoff 1963) versus placebo estimates using only pre-program cohorts with a fake cutoff at 1956 (red triangles). Placebo estimates are indistinguishable from zero across all subgroups, confirming the absence of pre-existing trends.

However, education is a ladder. A natural question is then whether INPRES effects continue beyond their direct target. If the program just reshuffled who completed primary school without changing subsequent trajectories, we would expect zero effects at higher levels (e.g., junior high school, high school, and university). If, however, primary completion was a genuine stepping stone, we should see positive effects further up the ladder.

We test this by redoing equation (1) identically to Table 1, but replacing the outcome variable with two alternative indicators: *secondary completion*, defined as completing at least junior secondary school (r_{jrhigh} or r_{jrvoc} , or any higher level), and *higher completion*, defined as completing at least senior secondary school (r_{srhigh} , r_{srvoc} , r_{coll} , or r_{uni}). All specifications, samples, fixed effects, trend controls, and clustering are identical to Table 1. We report results for all individuals, males, and females separately.

Table 3 presents the results. We found three patterns. First, effects decay monotonically up the education ladder. In Panel B, the pooled DiD coefficient falls from 0.032*** for primary to 0.025* for secondary and 0.012 (insignificant) for higher education. This gradient shows that the program’s direct impact was on primary completion. A fraction of those

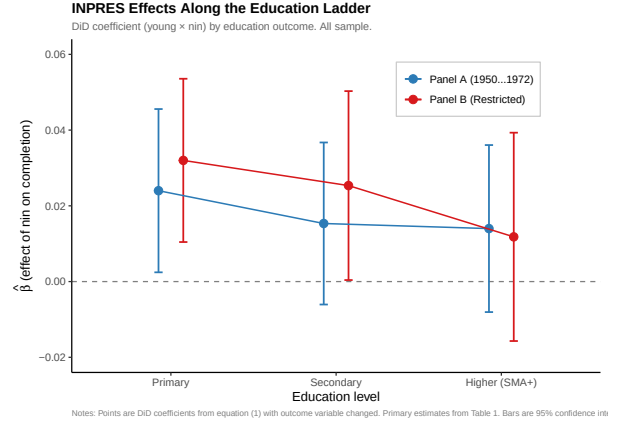


Figure 5: INPRES effects along the education ladder. Each point is the DiD coefficient $\hat{\beta}$ on $Exposed_t \times Inpres_d$ from equation (1) with the outcome variable changed to secondary and higher education completion. Bars are 95% confidence intervals. Both panels show monotonically declining effects: significant at primary, partially significant at secondary, and indistinguishable from zero at senior secondary and above.

additional completers continued to secondary school, and the effect dissipates further up the ladder where other constraints (cost, distance, ability) kick in.

Second, the secondary effect is marginally significant (Panel B, $p < 0.10$), suggesting the cascade is real, not noise. INPRES genuinely expanded the pool of students reaching junior secondary school.

Third, the female pattern from Table 1 persists. Female secondary coefficients (0.020 in Panel A, 0.028 in Panel B) exceed their male counterparts throughout, although they are statistically insignificant. By senior secondary and above effects for both sexes converge to zero, meaning the program’s reach did not extend this far.

Conclusion

This paper replicates and extends MRT’s[5] analysis of intergenerational human capital spillovers from Indonesia’s INPRES school construction program. Using the IFLS and district-level data from Duflo (2001) [3], we successfully replicate all qualitative findings from MRT’s two main tables. INPRES significantly raised primary school completion for directly exposed cohorts, with effects concentrated among women. This educational gain transmitted

Table 3: Extension 2: INPRES Effects Along the Education Ladder

	Primary (1)	Secondary (2)	Higher (SMA+) (3)
<i>Panel A: Full sample (1950–1972)</i>			
All	0.024* (0.011)	0.015 (0.011)	0.014 (0.011)
N	14,047	14,006	14,006
Male	0.020 (0.013)	0.008 (0.013)	0.014 (0.012)
N	7,099	7,076	7,076
Female	0.026* (0.014)	0.020 (0.013)	0.017 (0.014)
N	6,948	6,930	6,930
<i>Panel B: Restricted (1957–62 or 1968–72)</i>			
All	0.032*** (0.011)	0.025** (0.013)	0.012 (0.014)
N	7,779	7,752	7,752
Male	0.022 (0.015)	0.021 (0.014)	0.014 (0.016)
N	3,943	3,925	3,925
Female	0.040** (0.015)	0.028 (0.019)	0.006 (0.019)
N	3,836	3,827	3,827

Notes: Each cell reports the DiD coefficient $\hat{\beta}$ on $Exposed_t \times Inpres_d$ from equation (1), with the outcome variable changed. Primary = completed primary school. Secondary = completed junior secondary or above (`r_jrhigh`, `r_jrvoc`, or higher). Higher = completed senior secondary or above (`r_srhigh`, `r_srvoc`, `r_coll`, `r_uni`). All specifications the same as Table 1: district FE, birth-year FE, and birth-year FE interacted with `en71`, `ch71`, `wsppc`. Individual controls: `male`, Javanese ethnicity, missing-ethnicity indicator, birth-month FE. Standard errors clustered at district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

to the next generation through maternal, but not paternal, channels. The effect is improving children’s primary exam scores, with the largest effects for daughters of exposed mothers. The estimates are systematically below the paper’s, a discrepancy we trace to a difference in the construction of the INPRES intensity variable `nin`.

The first extension validates the parallel trends assumption underlying the DiD design. Using pre-program cohorts from the IFLS itself (a direct substitute for the unavailable SUPAS 1985 data used by MRT) we observe parallel pre-program trends between high- and low-intensity districts in primary school completion rates. An event study confirms this formally. Wald tests fail to reject joint nullity of pre-period coefficients in both the continuous

($F = 0.97$, $p = 0.48$) and binary ($F = 0.66$, $p = 0.79$) specifications. A pretrends power analysis following Roth (2022)[4] shows the test achieves 80% power against violations as small as 0.007 probability points per year. This confirms the null result is informative. A placebo test assigning a fake cutoff at 1956 yields coefficients indistinguishable from zero across all subgroups, ruling out pre-existing trends as an explanation for our main findings.

The second extension asks whether INPRES effects continue beyond primary schooling. Re-estimating Table 1 with secondary and higher education completion as outcomes, we find a monotonically declining gradient. The effect is largest and most significant at primary, partially significant at secondary (Panel B: $\hat{\beta} = 0.025$, $p < 0.05$), and indistinguishable from zero at senior secondary and above. This stepping-stone pattern is precisely what the program’s design predicts. INPRES built primary schools, and its reach attenuated at higher levels where other constraints bind. The female advantage stays across the ladder.

All these findings support a consistent narrative. A large-scale primary school construction program in a developing country like Indonesia can generate significant returns not only for the directly exposed generation but also for their children (especially through maternal channel). From extension 2 we also see that large-scale educational investments, even those targeting only primary schooling, may have intergenerational returns that extend well beyond their direct scope.

There are several directions we can do for future research. First, residential mobility between the birth district and the survey wave makes it difficult to clearly interpret differences by urban or rural status. Data that separately records both birth location and schooling location would allow researchers to test these differences more accurately. Second, the maternal transmission channel we observe could operate through several mechanisms (such as maternal time investment, better access to information, or changes

in household bargaining power) but the current data cannot distinguish between them. Future research using detailed time-use data or intra-household expenditure data could help identify which mechanism is behind the effect. That can hopefully provide clearer guidance for policies aimed at strengthening intergenerational returns.

spillovers: Indonesia’s school construction and its effects on the next generation. *AEA Papers and Proceedings*, 109:243–249.

Data Sources

The empirical analysis uses two data sources:

- (i) **Original replication package** (Mazumder, Rosales-Rueda & Triyana, 2019). Available at: <https://www.openicpsr.org/openicpsr/project/116476/version/V1/view>
- (ii) **Duflo (2001) INPRES district data** (`inpresdata.dta`). Available at: <https://github.com/droodman/Duflo2001/blob/main/inpresdata.dta>

References

- [1] Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. Columbia University Press.
- [2] Currie, J. and Moretti, E. (2003). Mother’s education and the intergenerational transmission of human capital: Evidence from college openings. *Quarterly Journal of Economics*, 118(4):1495–1532.
- [3] Duflo, E. (2001). Schooling and labor market consequences of school construction in Indonesia: Evidence from an unusual policy experiment. *American Economic Review*, 91(4):795–813.
- [4] Roth, J. (2022). Pre-test with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.
- [5] Mazumder, B., Rosales-Rueda, M., and Triyana, M. (2019). Intergenerational human capital